

GILT EDGE NORTH AGCM, AB, Canada

WMO: 717910

Lat: **53.0710N**Lon: **110.6211W**Elev: **2123**StdP: **13.60**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.6	-19.2	-31.1	1.0	-24.6	-25.1	1.5	-19.0	30.0	28.9	26.2	26.1	5.6	260	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	83.6	63.8	80.2	62.1	77.2	61.3	67.2	78.5	65.0	76.1	63.0	73.4	10.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	93.3	72.7	60.9	86.3	70.3	58.9	80.2	68.1	32.9	78.4	31.1	76.0	29.6	73.4	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.4	19.7	DB	-32.2	89.2	3.9	3.7	-35.0	91.8	-37.3	93.9	-39.5	96.0	-42.3	98.7	(4)
(5)			WB	-32.4	70.9	3.9	2.6	-35.2	72.8	-37.5	74.3	-39.7	75.7	-42.5	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	11.1	10.1	22.8	37.6	51.8	59.1	63.5	61.2	52.8	39.7	23.1	9.3
	DBStd	22.72	15.47	14.37	13.70	9.62	8.02	5.74	5.17	5.76	8.31	8.97	12.78	14.44
	HDD50	6100	1205	1118	843	382	75	8	0	1	64	337	806	1261
	HDD65	10316	1670	1538	1308	821	411	191	92	144	373	786	1256	1726
	CDD50	1353	0	0	0	11	132	280	418	348	148	16	0	0
	CDD65	92	0	0	0	0	3	13	45	25	6	0	0	0
	CDH74	1433	0	0	0	5	122	185	547	390	179	5	0	0
	CDH80	311	0	0	0	0	14	28	135	83	51	0	0	0
Wind	WSAvg	8.9	9.1	8.6	9.5	10.6	9.6	8.7	8.0	7.5	8.2	9.3	8.8	8.4
Precipitation	PrecAvg	16.4	0.7	0.5	0.6	1.1	1.7	2.9	3.1	2.3	1.4	0.7	0.6	0.7
	PrecMax	23.8	1.7	1.1	1.7	3.6	4.0	5.2	6.5	5.8	3.8	1.5	2.1	1.3
	PrecMin	8.8	0.2	0.2	0.2	0.1	0.2	0.6	0.9	0.4	0.1	0.1	0.1	0.3
	PrecStd	3.5	0.4	0.2	0.4	0.8	1.0	1.2	1.5	1.0	0.9	0.4	0.5	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	42.1	54.9	71.7	81.7	83.6	89.0	86.4	86.4	72.5	55.7	41.4
		MCWB	36.9	36.8	43.4	50.3	56.7	63.2	67.1	64.0	60.8	54.9	43.4	35.5
	2%	DB	39.6	37.7	48.7	65.1	77.8	78.9	83.6	81.8	79.2	63.2	47.5	37.7
		MCWB	35.0	33.8	40.4	47.1	55.4	60.5	66.7	63.3	58.5	50.2	39.9	33.3
	5%	DB	36.6	34.2	43.2	59.1	72.6	75.4	79.8	78.4	73.7	58.1	42.1	33.2
		MCWB	32.9	31.4	37.5	44.5	53.2	59.2	64.8	63.0	56.8	47.0	36.4	29.6
	10%	DB	32.8	31.5	39.0	53.9	68.5	71.9	76.4	74.9	68.3	53.6	38.2	28.9
		MCWB	29.9	29.0	34.5	41.8	51.2	57.7	63.3	61.4	54.5	44.6	34.0	26.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	37.3	45.1	51.2	58.8	65.8	71.4	68.9	62.9	55.7	45.3	36.4
		MCDB	41.7	41.7	54.1	70.9	77.1	77.5	83.6	79.7	80.0	70.3	53.1	40.8
	2%	WB	35.3	34.1	40.6	47.6	56.4	62.9	68.2	66.3	60.1	51.8	40.7	33.5
		MCDB	39.2	36.7	47.7	62.8	73.1	74.4	79.8	77.1	76.2	60.3	47.0	37.7
	5%	WB	33.0	31.6	37.6	45.0	54.5	61.0	66.2	64.1	57.8	48.3	37.1	29.8
		MCDB	36.4	34.2	43.3	57.5	69.7	71.2	77.2	75.0	71.6	56.9	41.2	33.0
	10%	WB	30.0	28.8	34.8	42.4	52.6	59.2	64.2	62.2	55.4	45.3	34.4	26.3
		MCDB	32.8	31.5	38.7	52.2	66.6	69.7	74.6	72.4	67.3	53.2	37.8	28.9
Mean Daily Temperature Range	5% DB	MDBR	16.2	16.8	16.6	19.6	23.3	20.5	21.3	21.8	22.6	18.7	14.2	14.0
		MCDBR	17.0	18.7	20.6	29.1	29.9	26.0	26.9	28.9	32.7	27.6	19.2	18.7
		MCWBR	13.9	15.7	13.8	14.8	13.3	11.6	12.3	13.3	14.7	15.4	12.9	15.6
	5% WB	MCDBR	17.1	17.4	19.9	27.3	27.3	22.7	24.3	24.9	30.0	25.9	18.0	18.2
		MCWBR	14.2	14.9	13.6	14.7	12.7	11.3	12.2	12.5	14.2	14.9	12.8	15.5
Clear-Sky Solar Irradiance	taub	0.246	0.294	0.284	0.330	0.352	0.371	0.364	0.376	0.314	0.278	0.251	0.225	
	taud	2.336	2.208	2.404	2.363	2.351	2.328	2.375	2.319	2.470	2.543	2.452	2.361	
	Ebn at Noon	243	255	290	287	285	279	279	268	276	267	242	233	
	Edh at Noon	18	28	29	34	36	38	36	36	27	20	16	14	
All-Sky Solar Radiation	RadAvg	327	662	1134	1496	1843	1889	1972	1606	1132	664	327	244	
	RadStd	32	49	88	135	127	134	103	124	121	91	34	23	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page